

# COMS 3003A

## Test 2

3 May, 2024

This is a closed book test.  
Time allowance: 60 minutes

1. (a) (10 points) Prove, using diagonalization, that the self-rejection problem for Turing machines is algorithmically unsolvable, i.e., that the language  $\{\langle M \rangle \mid M(\langle M \rangle) = 0\}$  is undecidable. You may not assume unsolvability of any other problem.  
(b) (10 points) Prove, using unsolvability of the problem from (a), that the problem of determining whether a Turing machine rejects an input word is unsolvable, i.e., that the language  $\{\langle M, w \rangle \mid M(w) = 0\}$  is undecidable.
2. (10 points) Prove that the problem of determining whether a Turing machine ever writes a 1 is algorithmically unsolvable. You may assume that the following problems are unsolvable:
  - self-acceptance for Turing machines;
  - acceptance for Turing machines;
  - halting for Turing machines.
3. (a) (5 points) Prove that the set of Turing machines is countable.  
(b) (10 points) Prove that the set of decision problems is uncountable.  
(c) (5 points) What conclusion can be drawn from (a) and (b)? Explain your answer.
4. The first-order language of arithmetic contains the constant 0, the unary function symbol  $S$  (successor), and the binary predicate symbols  $\cdot$  and  $+$ . Translate the following statements into this language (you are free to define any other symbols in terms of the ones listed above):
  - (a) (5 points) Every natural number is divisible by 1.
  - (b) (5 points) There is no biggest natural number.